

ThyroTrack: Holistic Endocrinal Cancer Prognosis Platform

Introduction

Thyroid nodules occur in approximately 67% of adults, yet only 5–15% progress to malignancy, creating a significant diagnostic challenge (Dean & Gharib, 2008). Current standards of care—including **ACR TI-RADS ultrasound scoring** and **fine-needle aspiration (FNA)**—are effective for clearly benign or malignant cases but often fail in **indeterminate nodules (Bethesda III–IV)**, leading to diagnostic uncertainty and unnecessary thyroidectomies with associated costs and morbidity (Haugen et al., 2016; Onenerk et al., 2017). Although recent artificial intelligence approaches have improved ultrasound-based risk prediction, most models remain **static**, **binary classifiers**, neglecting longitudinal disease progression, patient history, and inter-patient similarity (Cao et al., 2023; Ajilisa et al., 2023). To address these limitations, this project introduces **ThyroTrack**, a multimodal deep-learning framework integrating a **CNN** (convolutional neural network) for imaging biomarkers with 3 ensemble classical models; **GRU** (gated recurrent unit) for temporal disease modeling; and **GNN** (graph neural network) for patient similarity analysis, enabling interpretable, longitudinal thyroid cancer risk stratification using combined imaging and clinical data.

Rationale/Engineering Goal

- 75% of “suspicious” thyroid surgeries are benign (Mavromati et al., 2023)
- TI-RADS and FNA do not capture:
- Longitudinal hormone trajectories
 - ◆ Disease evolution over time
 - ◆ Inter-patient similarity patterns
- Develop an integrated CNN–GRU–GNN framework that achieves $\geq 90\%$ multiclass AUC for thyroid nodule characterization and reduces unnecessary surgeries by $\geq 30\%$ compared to TI-RADS guidelines.

Design & Data Characterization

Figure 1: Clinical diagnostic workflow and cytology-based decision tree for thyroid nodules.

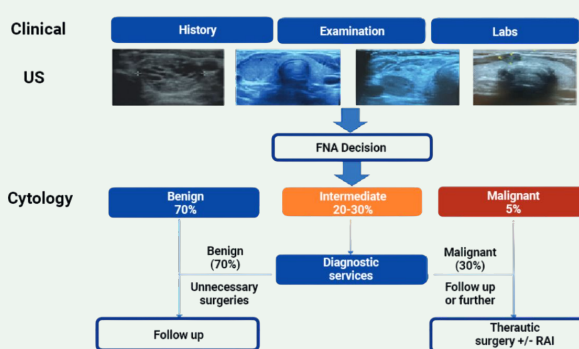
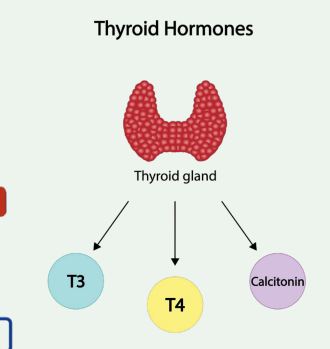


Figure 2: Illustration of the thyroid gland & primary hormonal secretions.



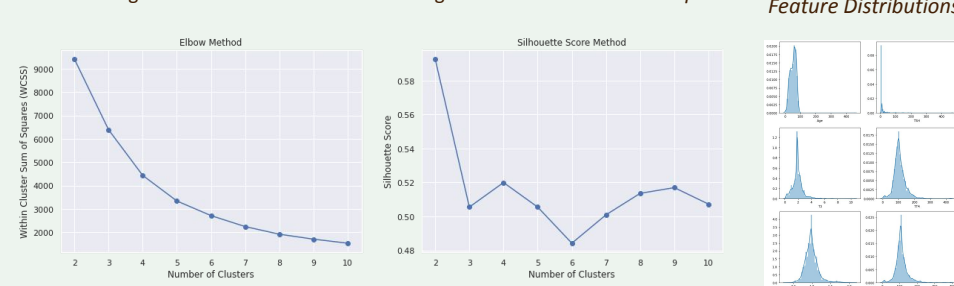
Datasets

- Clinical/Laboratory data: 3,772 patient records, 30 endocrine (i.e. TT4, FTI, T3, T4U, TSH, etc.) and demographic features (i.e. sex, age, etc.)
- Imaging data: ~500 thyroid ultrasound images (DDTI dataset), TI-RADS
- Exploratory Clustering Analysis (Pre-Modeling)**
- Purpose: identify latent endocrine structure prior to supervised learning
- Features used: Standardized TT4 and FTI with K-Means clustering (k=4):
 - ◆ Within-Cluster Sum of Squares (Elbow Method)
 - ◆ Silhouette Score (0.40–0.65 range) confirming distinct separability of Sick, Hyper, Hypo, & Negative endocrine states

Figure 3: Elbow Curve

Figure 4: Silhouette Score Graphs

Figure 5: Patient EHR Feature Distributions



Pipeline Procedures

GRU–GNN Hybrid Noninvasive Branch (Clinical + Temporal):

- GRU (Temporal Modeling):
 - Sequential hormone modeling; 32-dimension embedding.
- GNN (Relational Modeling):
 - Nodes = individual patients
 - Edges = demographic + laboratory feature similarity
- 80% training / 20% testing; Adam optimization, adaptive LR schedule.

Figure 6: End-to-End Diagnostic / Classification Methodology Pipeline with Ultrasound Segmentation Masks + Localizations.

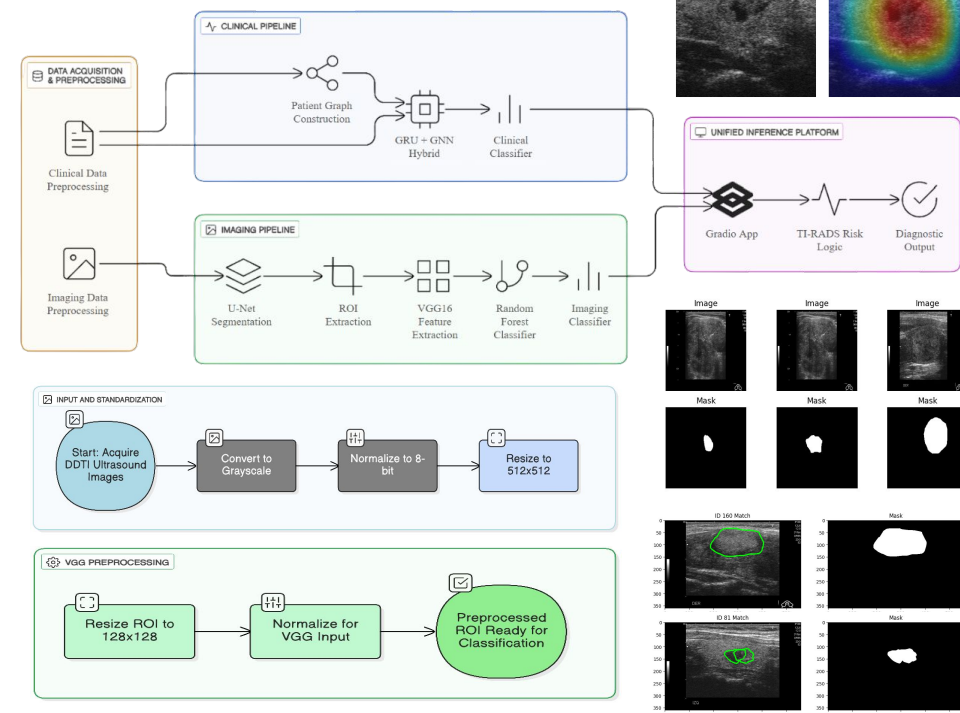


Figure 7: GradCAM Saliency Heatmaps - Benign, Suspicious, Malignant Cases

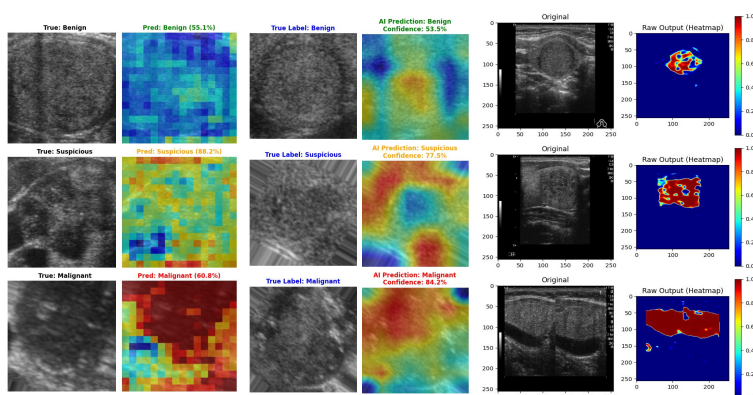
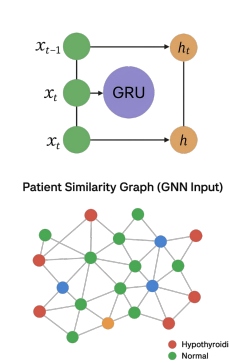


Figure 8: GRU and GNN Input Data Structures



Statistical Analysis

Table 1: AOC Performance Compared to Prior Detection Methods (Buda et al., 2019)

Methods	Class Types	ROC AUC
CNN	SVM (0-4)	0.96
	Random Forest (0-4)	0.97
GRU-GNN	Negative	0.84
	Hyper	0.98
	Hypo	0.91
	Sick	0.90
Prior study - Deep learning	Binary	0.87
Prior study - ACR TI-RADS committee consensus	Binary	0.91
Prior study - Radiologists (mean)	Binary	0.83

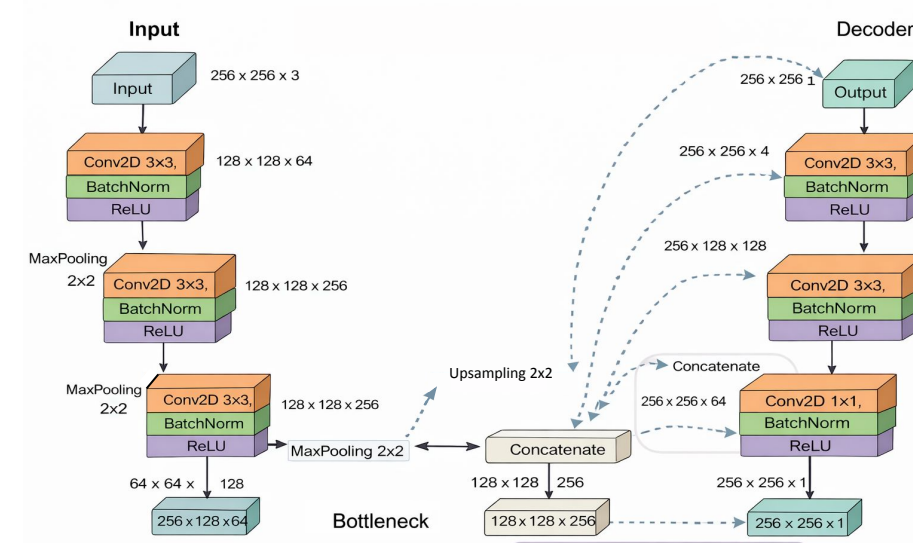
- **McNemar’s test** showed a statistically significant improvement in classification accuracy after GRU-GNN optimization
 - $\chi^2(1) = 13.30$
 - $p = 0.00027$
- **DeLong’s test** demonstrated that the GRU-GNN model significantly outperformed existing standards
 - vs. radiologists (mean): $p = 0.00065$
 - vs. prior deep-learning models: $p = 0.0117$
 - vs. ACR TI-RADS consensus guidelines: $p = 0.0394$

Classical Machine Learning Classifiers

CNN Branch (Imaging Feature Extraction) - VGG16 Backbone:

- Support Vector Machine (SVM)
 - Kernel: Radial Basis Function (RBF)
 - Probability calibration enabled
 - Class weighting applied to mitigate residual imbalance
- Random Forest (RF)
 - Number of trees: 200
 - Bootstrap sampling enabled
 - Gini impurity for split criterion
- Ensemble Strategy
 - Soft-voting ensemble combining SVM and RF
 - Final class probability = mean of individual classifier probabilities

Figure 9: Schematic architecture of the U-Net CNN (Encoder, Bottleneck, Decoder)



ThyroTrack Results

Figure 10: Confusion Matrix of Final GNN-GRU Classification Accuracies

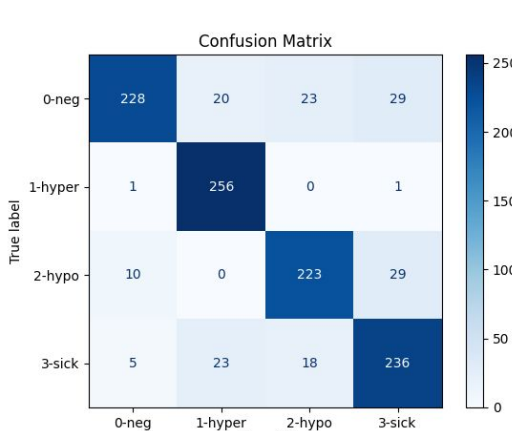


Table 2: Statistical Metrics of ThyroTrack Modular Performance

Metrics	Score
Accuracy	0.860 ± 0.018
Precision	0.861 ± 0.017
Recall	0.858 ± 0.019
F1 Score	0.859 ± 0.016
AUC Score	0.910 ± 0.012
Specificity	0.852

Figure 11: ROC AUC Curve of Final GNN-GRU Classification Labels

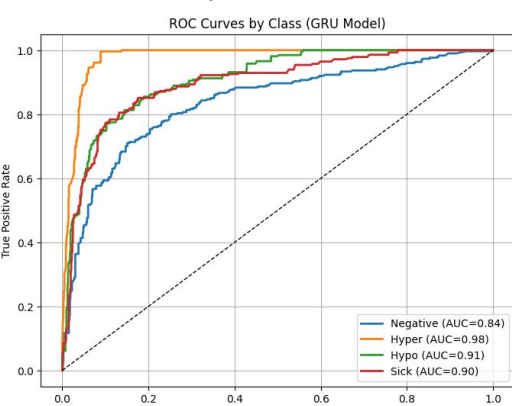


Figure 12: Correlation Matrix with Heatmap for Thyroid Feature Metrics



Figure 13: Feature Importance from Logistic Regression Baseline Bar Chart

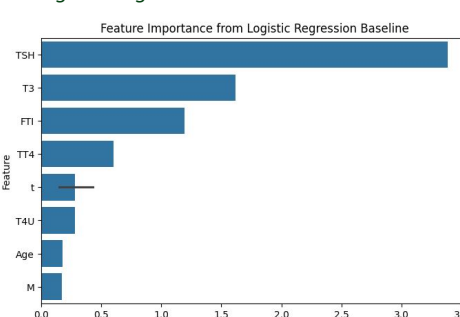


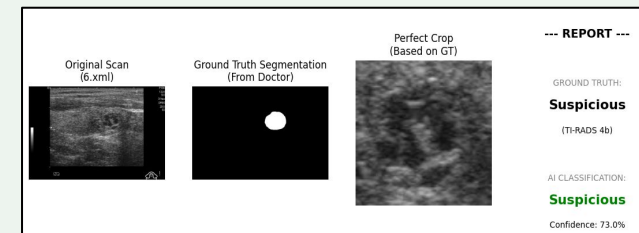
Table 3: Model Predictions of New Data from ThyroTrack Randomized Sample

	True Label Sample	Predicted Label Sample
Subject 1	3	3
Subject 2	1	1
Subject 3	0	0
Subject 4	0	3
Subject 5	3	3
Subject 6	2	2
Subject 7	1	1
Subject 8	2	2
Subject 9	1	1
Subject 10	2	2

Discussions/Platform

- TSH identified as **most critical prognostic feature** (coefficient >3.5).
- **Unified Gradio inference interface** enables secure cloud-based deployment with Firebase Authentication.
- GRU model achieved multiclass **AUC ≥ 0.90** across endocrine states (Hyper: 0.98, Hypo: 0.91, Sick: 0.90).

Figure 14: AI Diagnostic Workflow and Classification Result



Conclusions

- Achieved a **macro-averaged ROC-AUC of 0.91** across four endocrine classes, with $\sim 0.90\%$ accuracy, meeting the engineering goal.
- ◆ High AUC scores (≥ 0.90) suggest potential to **reduce diagnostic uncertainty** and **guide follow-up intervals**.
- GRU-based temporal modeling proved effective for **longitudinal disease trajectory prediction**, previously unexplored in ultrasound.
- Multiclass architecture successfully differentiated **5 distinct thyroid conditions** (Normal, Goiter, Thyroiditis, Adenoma, Carcinoma) with highest accuracy on Carcinoma (189/total) & Normal (182/total) cases
- **CNN sensitivity-specificity trade-off** requires tuning; full GNN integration essential before real-world deployment.
- Prospective validation on **independent cohorts** needed to assess real-world generalizability and clinical utility.

Applications/Future Studies

- Potential **adjunctive decision-support tool** deployable in routine ultrasound workflows via **cloud-based platform**.
- Framework applicable to other **endocrine disorders and organ systems** with imaging and longitudinal EHR data.
- **Multi-center prospective validation** across different ultrasound machines, operators, and patient populations.
- Integration and optimization of GNN component for **improved risk stratification** in indeterminate nodules.
 - ◆ Targeted to resolve **51+55 misclassifications**
- Incorporate **additional biomarkers** (calcitonin, thyroglobulin) to refine risk stratification for aggressive thyroid cancers.
- **Expand NLP pipeline** to extract clinical context from radiology reports for improved trajectory forecasting.

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